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Faculty of Informatics and Information Technologies

NAO ROBOT EXERCISE SYSTEM

Backlog Management Methodology

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1 Purpose of the Document

This document describes how the NAOMI team manages its project backlog using Jira and a Scrumban approach. The aim is to make planned work visible, keep responsibilities clear, and support regular delivery of results throughout the academic year.

Scrumban combines sprint-based planning and review with the visual workflow of a Kanban board. For NAOMI, this means agreeing on short-term goals while allowing the team to monitor work in progress, identify blockers, and adjust priorities when necessary.

2 Scope and Responsibilities

The methodology applies to all NAOMI team members working on analysis, robot-control software, motion processing, testing, documentation, and other project activities. Jira serves as the central place for recording and following the status of these activities.

Every member is expected to keep their assigned issues up to date. The team jointly reviews priorities, dependencies, and progress. Coordination responsibilities are assigned to a team leader.

3 Backlog Organisation in Jira

3.1 Product Backlog

The product backlog contains the features, improvements, research tasks, experiments, and technical work that may be needed to reach the project's goals. Items should have a concise title and a description of the expected outcome. Larger activities may be represented as Epics and divided into smaller Stories or Tasks, using the issue types available in the team's Jira project.

The backlog is reviewed and reordered when requirements change or new information becomes available. Higher-priority items should be sufficiently clear before the team starts working on them.

3.2 Sprint Backlog

At the beginning of each sprint, the team selects a realistic set of issues from the product backlog, taking into account available capacity, priorities, and dependencies. The selected work is visible on the Jira board. Unfinished issues are discussed during the sprint review or next planning session and are not automatically treated as completed.

4 Scrumban Workflow

The NAOMI team uses a Jira board to manage tasks and track their progress throughout the development process. Our workflow consists of five stages, allowing team members to monitor the current state of each issue.

Table 1: Scrumban workflow states

Status	Description
To Do	The task is planned but work has not started.
In Progress	A team member is actively working on the task.
In Review	The implementation is completed and waiting for review or feedback from other team members.
Testing	The task has passed the review stage and is being tested to verify that it works correctly and meets the requirements.
Done	The task has been reviewed, tested, and successfully completed.

Team members are responsible for keeping the status of their assigned issues up to date in Jira. Tasks should be moved between columns as soon as their status changes, rather than only at the end of a sprint.

If a task cannot be completed due to a blocker, the responsible team member should document the problem in Jira and inform the rest of the team.

Following the Scrumban approach, the team focuses on completing tasks already in progress before starting new ones. This helps avoid unnecessary multitasking and improves the overall workflow.

5 Reviewing Progress and Identifying Blockers

The Jira board is checked during regular team meetings and whenever an assignee makes meaningful progress. During a review, the team considers whether the cur-

rent sprint goal is still achievable and whether work is moving through the board as expected.

- **Priorities:** Are we working on the most important issues? Have new requirements changed the plan?
- **Progress:** Which issues have remained In Progress for too long? Is anything blocked or waiting for a decision?
- **Ownership:** Does each active issue have a clear assignee? Are issue descriptions and next steps understandable?
- **Delivery:** Are completed results tested, documented, and ready to demonstrate or integrate?

Issues that need support should be raised at the next team meeting or communicated earlier through the team's agreed communication channel. Critical blockers should not be left unresolved until the sprint ends.

6 Rules for Managing Jira Issues

6.1 Creating and Updating Issues

An issue should describe a specific activity or expected result. It should normally include a title, a short description, an assignee when assigned, and relevant acceptance criteria. Links to code, designs, experiments, or documentation may be attached where useful.

Each member updates the status of their issues and records important findings or blockers in Jira. If an issue is too broad or cannot be completed within a reasonable period, it should be divided into smaller pieces of work.

6.2 Changes During a Sprint

Scrumban allows the team to react to new information without losing visibility of its commitments. Urgent work may be added during a sprint after discussion with the team, particularly if it affects the sprint goal or another member's progress. The reason for a significant change should be documented in the related Jira issue.

6.3 Finishing Work

An issue is moved to Done only when its agreed acceptance criteria are satisfied. Depending on the type of work, this may involve a code review, a successful test, an updated document, or a demonstration of the result. The team should use its Definition of Done, once agreed, as the common reference for deciding whether an item is finished.

7 Planning and Estimation

The team estimates work when an estimate is useful for sprint planning and prioritisation. Story points may be used to express relative effort, complexity, and uncertainty. An issue type does not determine a fixed number of story points: for example, different Tasks or Stories can require very different amounts of work.

Table 2: Illustrative relative effort scale for discussion

Points	Interpretation
1	Very small, well-understood task
2	Small task with limited uncertainty
3	Moderate task requiring several steps
5	Larger task involving more effort or uncertainty
8	Complex activity that may need to be split

This scale is a **proposal**, not an already approved NAOMI rule. The team can adopt or change it after discussing a few example issues. Capacity estimates should also account for university deadlines and the availability of individual members. At the end of a sprint, completed work, carried-over tasks, and blockers can be used to improve the next planning session.

8 Continuous Improvement

At the end of each sprint, the team briefly reviews what worked well, what slowed progress, and which changes would help in the next sprint. Possible improvements include clearer Jira issue descriptions, more consistent board updates, smaller tasks, and adjustments to meeting or review practices.

The purpose of this methodology is not to introduce unnecessary administration. Jira

should give the team a reliable overview of its work while leaving enough time for actual development and experimentation.